CS486 MOCK FINAL EXAMINATION - TEST 1

Based on the FINAL REVIEW structure; total: 10 points

Question 1 - ER Models [1 pt]

A university sports center manages **Clubs**, **Teams**, **Athletes**, and **Coaches**. Club names are unique. Each club has one or more teams; team names are unique only within a club. Each athlete belongs to exactly one team and has a jersey number that is unique only within that team. A coach may coach several teams, and a team may have several coaches. One coach on each team is designated as the head coach.

Draw an ER diagram showing keys, participation, and cardinalities. Then state the primary key that would naturally identify each of Club, Team, and Athlete after mapping to relations.

Question 2 - Query Languages [2 pt]

Consider:

$$Student(SID, Name, Major, YearLevel)$$
$$Course(CID, Title, Dept, Credits)$$
$$Enroll(SID, CID, Semester, Year, Grade)$$
$$Prerequisite(CID, PreCID)$$

Write:

(a) a relational algebra expression to find students who have taken *every* prerequisite of course 'DB'.

(b) a tuple relational calculus expression to find students who enrolled in at least one course offered by their own major/department.

(c) one concise SQL query to return, for each major, the student(s) with the highest number of distinct courses passed (Grade ≥ 5). Ties must be kept.

Question 3 - Database Constraints [1 pt]

Schemas:

$Employee(EID, Name, Dept, Salary)$, $Project(PID, Dept, Budget)$, $WorksOn(EID, PID, Hours)$.

Business rule: *An employee may work on a project only if the employee and the project belong to the same department; additionally, Hours must be positive and at most 40 for each employee-project pair.*

Express the cross-table rule formally in TRC or RA, and show an SQL implementation using appropriate declarative constraints and/or a trigger/pseudocode.

Question 4 - Serialization & Concurrency [2 pt]

Given

$$S = r_1(A)\ w_2(A)\ r_3(B)\ w_1(B)\ r_2(B)\ w_3(A)\ w_2(C)\ r_1(C).$$

(a) Construct the precedence graph and determine whether S is conflict-serializable. If yes, list all conflict-equivalent serial orders; if not, identify a cycle.

(b) Determine whether S is recoverable and cascadeless if $w_i(X)$ means the written value becomes visible immediately and commits occur in the order c_3, c_1, c_2 after the shown operations.

(c) Briefly explain whether strict 2PL would permit this exact interleaving.

Question 5 - Recovery [1 pt]

A system uses WAL and an **UNDO/REDO** immediate-update protocol. The log at crash time is:

```
<start ckpt (T1,T2)>
<T1, A, 10, 14>
<T2, B, 20, 25>
<commit T1>
<T3, C, 30, 36>
<end ckpt>
<T2, D, 40, 44>
<T3, C, 36, 38>
<abort T2>
--- crash ---
```

State which transactions are winners/losers and give the REDO and UNDO actions required for recovery.

Question 6 - Functional Dependencies & Normalization [1 pt]

Let $R(A, B, C, D, E, F)$ with

$$F = \{A \to BC,\ C \to D,\ D \to E,\ BE \to F,\ F \to A\}.$$

(a) Compute A^+ and determine whether A is a candidate key.

(b) Determine the highest normal form satisfied by R (1NF/2NF/3NF/BCNF), with a brief reason.

(c) Is the decomposition $R_1(A, B, C), R_2(C, D, E), R_3(B, E, F)$ guaranteed lossless? Justify briefly.

Question 7 - Concurrency Control Execution [1 pt]

Initially $X = 100$. Two transactions execute without isolation:

```
T1: read(X); X := X - 20; write(X)
T2: read(X); X := X * 1.10; write(X)
```

List all distinct final values of X that can result from interleavings preserving each transaction's internal order. Identify the anomaly for any result that cannot occur under either serial order.

Question 8 - Query Optimization [1 pt]

A relation Orders has 1,000,000 tuples, 100 tuples/page, hence 10,000 data pages. Attribute OrderID is unique; Status has 10 uniformly distributed values. Assume:

- a clustered B+ tree on OrderID of height 3;
- an unclustered B+ tree on Status of height 3;
- a static hash index on OrderID with expected 2 bucket I/Os.

Estimate page I/O cost for (i) equality on OrderID using B+ tree vs hash, and (ii) retrieving all rows with one Status value using the Status index vs full scan. Which access method should the optimizer prefer in each case? State assumptions.

– End of Mock Test –